Space Emanation Theory

Capacity, Common Energy and Orbital Dynamics

Rough draft 1

18 October 2026

Abstract

Space Emanation Theory (SET) describes gravitational burden, translation and clock progression through one capacity relation. This paper asks what follows when that clock progression must be generated by the same energy that governs motion and the complete source. An actual internal clock supplies $\partial\mathcal{H}/\partial E_0 = \beta$. With regular canonical mechanics and test-body additivity, this selects the probe Hamiltonian and its reduced action. Internal energy then contributes equally to local inertia and passive weight. The requirement is extended to interacting sources, field energy and the stresses supporting a stationary configuration. Within a stated local static energy class, complete-energy consistency and the SET source law at physical spherical source surfaces select the clock and radial responses together. Their field equations yield a conserved vacuum work current. The spherical orbit equation and its perihelion advance follow without choosing an orbital correction coefficient. A further extension to complete-system bulk motion selects the leading mixed-momentum interaction and its stationary rotational response. The resulting spherical and leading two-body formulas agree with established relativistic results. The contribution is the derivation through capacity-governed events, a common energy and the SET source law, with the class of admitted responses stated at each selection.

1 The physical question

A clock participates in the field it is used to measure. Its ticks are physical events. If the field limits their progression, the energy law that generates those events must respect the same limit as the energy law that moves the clock. SET takes this requirement seriously for the source as well as the probe: the material, its support and its field belong to one physical system.

The starting relation is

$$S^2 + V_{\mathrm{rel}}^2 + c^2\beta^2 = c^2, \qquad \mathrm{d}\tau = \beta\,\mathrm{d}T. \tag{1.1}$$

Here S is gravitational burden expressed in speed units, V_{rel} is the translational motion reading, and β is the body's clock progression relative to reference time T. In the isolated stationary description used below, T is calibrated in the unloaded exterior at infinity. For a held clock, write $\alpha = \beta$, so

$$A \equiv \alpha^2 = 1 - \frac{S^2}{c^2}. \tag{1.2}$$

The investigation turns this physical interpretation into a requirement on energy derivatives. It then asks whether the same requirement can constrain the source response. The central result is

a connected construction:

$$\begin{aligned} \text{capacity-governed events} &\longrightarrow \text{clock-consistent probe energy},\\ \text{complete source energy + SET source law} &\longrightarrow \text{static response and work current},\\ \text{probe energy + selected source response} &\longrightarrow \text{orbital dynamics}. \end{aligned} \tag{1.3}$$

The first selection uses regular canonical mechanics and ordinary addition of test-body energy. The static field selection uses a local isotropic response with quadratic first spatial derivatives. These are the mechanical and field classes examined here. Stating them makes the selection precise: alternative coefficients inside those classes can be followed to their consequences and tested against the same physical requirements.

Throughout, $\mathcal{H}$ denotes a probe Hamiltonian; H denotes the complete rest-frame energy of an isolated source. A coordinate spatial derivative is $\boldsymbol{\nabla}$; $\boldsymbol{D}$ uses physical spatial lengths. The work current $\boldsymbol{\mathcal{J}}$, introduced below, has acceleration units. It is distinct from angular momentum.

2 From the capacity identity to energy and distance

Multiplying Eq. (1.1) by $m/2$ and defining $K_0 = mV_{\rm rel}^2/2$ gives

$$K_0 = \frac{mc^2}{2}(1-\beta^2) - \frac{mS^2}{2}, \tag{2.1}$$

$$\Delta K_0 = -\frac{m}{2}\Delta(S^2) - \frac{mc^2}{2}\Delta(\beta^2), \tag{2.2}$$

$$\mathrm{d}K_0 = -mS\,\mathrm{d}S - mc^2\beta\,\mathrm{d}\beta. \tag{2.3}$$

Thus an increase in burden and translation is balanced by a decrease in the clock contribution. The identity does not require K_0 itself to remain constant. The physical conserved energy will be obtained from the evolution law.

The corresponding distance identity uses two powers of the time increment:

$$\mathrm{d}\ell^2 = V_{\rm rel}^2\mathrm{d}T^2 = c^2(\mathrm{d}T^2 - \mathrm{d}\tau^2) - S^2\mathrm{d}T^2. \tag{2.4}$$

For a varying field, the travelled operational distance is

$$\ell = \int \sqrt{c^2(1-\beta^2) - S^2}\,\mathrm{d}T. \tag{2.5}$$

Multiplying an energy by one power of time instead gives action units. Keeping these operations separate preserves the physical meaning of the capacity terms.

Let $v^i = \mathrm{d}x^i/\mathrm{d}T$ and introduce a positive spatial response q_{ij}:

$$V_{\rm rel}^2 = q_{ij}v^iv^j, \qquad \mathrm{d}\ell^2 = q_{ij}\mathrm{d}x^i\mathrm{d}x^j, \qquad \beta^2 = A - \frac{q_{ij}v^iv^j}{c^2}. \tag{2.6}$$

This describes operational distances. A propagation-delay interpretation must reproduce these same readings; the notation does not decide a microscopic stretching mechanism. The probe derivation first treats A, q_{ij} as supplied fields. Their spherical selection is performed in Sections 6–7.

3 An actual clock selects the probe coupling

3.1 The clock is generated by energy

Give the body an internal periodic degree of freedom with phase ϑ, conjugate action I, and calibrated local energy $h(I)$. Its proper phase rate is h_I. Add this energy to the body's total rest energy:

$$E_0 = E_{\text{base}} + h(I), \qquad \mathcal{H} = \mathcal{H}(x, p, E_0). \tag{3.1}$$

The same regular Hamiltonian generates translation and the internal phase:

$$v^i = \mathcal{H}_{p_i}, \qquad \dot{\vartheta} = \mathcal{H}_{E_0} h_I. \tag{3.2}$$

SET assigns those actual clock events the rate $\dot{\vartheta} = \beta h_I$. For a functioning clock, $h_I \neq 0$, and therefore

$$\boxed{\mathcal{H}_{E_0} = \beta.} \tag{3.3}$$

The derivative holds position and canonical momentum fixed. It compares nearby internal energy states; it does not assert that the body's rest energy changes during its orbit. The clock frequency cancels and supplies no adjustable orbital parameter.

The invariant is now a condition on the unknown energy:

$$\boxed{S^2 + q_{ij}\mathcal{H}_{p_i}\mathcal{H}_{p_j} + c^2\mathcal{H}_{E_0}^2 = c^2.} \tag{3.4}$$

The clock meaning of capacity has entered the generator of evolution.

3.2 Additivity fixes the regular massive branch

For noninteracting identical comoving probes in a fixed background, use classical energy–momentum additivity:

$$\mathcal{H}(x, \lambda p, \lambda E_0) = \lambda \mathcal{H}(x, p, E_0), \quad \lambda > 0. \tag{3.5}$$

This is a test-body statement; changes of the background due to the probe are excluded. Euler's identity gives

$$\mathcal{H} = E_0\mathcal{H}_{E_0} + p_i\mathcal{H}_{p_i} = E_0\beta + p_i v^i. \tag{3.6}$$

Differentiate with respect to p_j, cancel the direct velocity terms, and use the dependence of β on v^i:

$$\frac{\partial v^i}{\partial p_j}\left(p_i + E_0\frac{\partial \beta}{\partial v^i}\right) = 0. \tag{3.7}$$

On the regular massive branch the momentum Hessian is invertible. Since $\partial\beta/\partial v^i = -q_{ij}v^j/(c^2\beta)$,

$$p_i = \frac{E_0 q_{ij} v^j}{c^2\beta}, \qquad \boxed{\mathcal{H} = \frac{E_0 A}{\beta} = \alpha\sqrt{E_0^2 + c^2 q^{ij}p_i p_j}.} \tag{3.8}$$

Its Legendre representation is

$$\boxed{L = p_i v^i - \mathcal{H} = -E_0\beta.} \tag{3.9}$$

The full phase-space action also contains $I\dot{\vartheta}$. For a cyclic internal phase, I is conserved. Reducing the motion problem at fixed I removes that phase term while retaining its energy in E_0. Equation (3.9) is the resulting probe coupling.

The action has therefore been obtained from actual clock progression, canonical evolution, additivity and regularity. It was not chosen by comparing candidate actions with Mercury's orbit. Appendix C explains why additivity and the field class do real work in the selection.

3.3 Inertia, passive weight and the weak limit

A held local observer uses $\mathrm{d}t_h = \alpha \mathrm{d}T$. If u is the probe's speed in that observer's orthonormal frame, then

$$V_{\mathrm{rel}} = \alpha u, \qquad y \equiv \frac{\beta}{\alpha} = \sqrt{1 - u^2/c^2}, \qquad p_{\mathrm{local}} = \frac{E_0 u}{c^2 y}. \tag{3.10}$$

Defining local rest inertia by momentum response gives

$$m_{\mathrm{i}} \equiv \left.\frac{\partial p_{\mathrm{local}}}{\partial u}\right|_{u=0} = \frac{E_0}{c^2}. \tag{3.11}$$

Thus $E_0 = mc^2$ follows after the coupling is selected. Expanding the same energy in a weak field at low speed gives

$$\mathcal{H} - E_0 = \frac{mV_{\mathrm{rel}}^2}{2} - \frac{mS^2}{2} + \text{higher-order terms}. \tag{3.12}$$

The same coefficient supplies inertia and the passive gravitational coupling.

There is also a shorter result for changes of internal energy, independent of selecting the whole baseline coupling. Partially Legendre-transform the center motion, $\mathcal{R} = p_i v^i - \mathcal{H}$, while retaining I. At fixed x, v, the momentum-derivative terms cancel:

$$\mathcal{R}_I = -\mathcal{H}_I = -\beta h_I, \qquad \Delta\mathcal{R} = -\beta \Delta h. \tag{3.13}$$

In the test background $\beta(x, v)$ is independent of the internal state. The resulting increments are

$$\Delta m_{\mathrm{i}} = \frac{\Delta h}{c^2}, \qquad \Delta \boldsymbol{F}_{\mathrm{hold}} = \frac{\Delta h}{c^2}\, c^2 \boldsymbol{D} \ln \alpha. \tag{3.14}$$

The force is the locally measured external force holding the body fixed. Its gravitational factor is obtained from clocks and distance, rather than defined by dividing a measured weight by mass.

4 Interacting clocks must belong to one complete energy

For interacting sources with independent internal energy parameters E_i, impose the same clock condition on one common energy:

$$H_{E_i} = \beta_i. \tag{4.1}$$

Smoothness then requires

$$\boxed{\frac{\partial \beta_i}{\partial E_j} = \frac{\partial \beta_j}{\partial E_i}.} \tag{4.2}$$

Changing one source's energy and then another's cannot give a different final energy merely because the order was reversed, when the final *complete state* is the same. Fields and supports must be included in that state. Different histories can leave different field states; Section 12 returns to that distinction.

4.1 Why complete energy is the source charge

Let z represent all internal canonical variables of a compact system, including its field and supports. Extend clock universality to this complete system. Under a spatially uniform external clock factor a, all its internal rates acquire that factor:

$$\partial_z H(a, z) = a\, \partial_z H(1, z). \tag{4.3}$$

Integration and an empty-state energy calibration give

$$H(a,z) = aH(1,z). \tag{4.4}$$

Place a distant weak probe of proper energy E_B at separation d. Write the leading conservative interaction with source A as

$$H_{AB} = H_A(z) + E_B - \frac{GE_BC_A(z)}{dc^4} + \text{higher-order and tidal terms}, \tag{4.5}$$

where C_A is the source charge in energy units. The external clock factor is $a = 1 - GE_B/(dc^4)$ to this order. Comparing all internal rates from Eqs. (4.4) and (4.5) requires $\partial_z C_A = \partial_z H_A$. The empty-source calibration therefore gives

$$\boxed{C_A = H_A, \qquad M_{\text{exterior}} = H_A/c^2.} \tag{4.6}$$

This uses a compact isolated subsystem and a weak distant interaction; the source's internal binding need not be weak. It is an operational extension of the clock requirement to the complete state.

The source mass of a strongly bound system consequently includes internal, support and binding energy. In weak binding it approaches the sum of material rest masses. Assigning complete rest energy the same unloaded bulk law as the free probe, $H_{\text{bulk}} = \sqrt{H_A^2 + c^2P^2}$, gives the same mass H_A/c^2 as its low-speed bulk inertia. The bulk extension will be tested more specifically in Section 10.

5 The source law and an exact assembly example

At a uniformly loaded physical spherical source surface, retain SET's law in its two readings:

$$Q_{\text{s}}^2 = 24\pi GMV_{\text{ref}} = 32\pi^2 GMR_{\text{s}}^3, \qquad Q_{\text{s}} = \mathcal{A}_s S_s, \tag{5.1}$$

where

$$\mathcal{A}_s = 4\pi R_{\text{s}}^2, \qquad V_{\text{ref}} = \frac{4\pi R_{\text{s}}^3}{3}, \qquad M = H/c^2. \tag{5.2}$$

The reference volume is defined by the areal radius; it is not automatically the integrated proper material volume. Combining the relations gives

$$S_s^2 = \frac{2GH}{R_{\text{s}}c^2}, \qquad \alpha_s^2 = 1 - \frac{2GH}{R_{\text{s}}c^4}. \tag{5.3}$$

Here the law is applied at actual source surfaces. The exterior field between reading surfaces will be obtained from the field equations.

Consider a supported thin shell at fixed physical radius R, with total proper material energy E, including the material carrying its stress. Put $g = G/(Rc^4)$. The source clock relation and actual-clock condition give

$$\frac{\partial H}{\partial E} = \sqrt{1 - 2gH}, \qquad H(0,R) = 0. \tag{5.4}$$

Integrating at fixed R, on the positive-clock branch, yields

$$\boxed{H = E - \frac{gE^2}{2}, \qquad \alpha_s = 1 - gE, \qquad gE < 1.} \tag{5.5}$$

The assembled energy is $H = \int_0^E \alpha_s(E')\,\mathrm{d}E'$. The source clock changes during assembly, so multiplying every element by the final clock rate would give a different answer.

Let the shell's surface pressure be P_s, with $\mathrm{d}E = -P_s \mathrm{d}\mathcal{A}_s$. Requiring stationary H under a radial variation gives

$$P_s = \frac{GE^2}{16\pi R^3 c^4 \alpha_s}, \qquad \boxed{H = \alpha_s(E + 2P_s\mathcal{A}_s).} \tag{5.6}$$

The pressure term follows from physical work and supplies the active contribution of the support.

For example, if $gE = 0.2$, then $H = 0.9E$ and $\alpha_s = 0.8$. The selected field construction below gives an exterior field contribution $0.1E$, so $H = 0.8E + 0.1E$. In the source-current integral the clock-weighted stress contributes the same $0.1E$. These are consistent decompositions of one system, rather than separate corrections.

An inner system of complete energy H_0 can also be included:

$$H = H_0 + E\sqrt{1 - 2gH_0} - \frac{gE^2}{2}. \tag{5.7}$$

With $\alpha_- = \sqrt{1 - 2gH_0}$ and $\alpha_+ = \alpha_- - gE$, $H_E = \alpha_+$ and $H_{H_0} = \alpha_+/\alpha_-$. Repeated assembly therefore gives nested sources whose individual clocks are derivatives of the same outer energy.

6 A common static energy with an unknown spatial response

The shell example exposes what the field construction must account for. We now allow both its clock and spatial responses to vary. In isotropic coordinates introduce positive fields ψ, ζ and unknown functions F, U:

$$N \equiv \alpha = \frac{\zeta}{F(\psi)}, \qquad q_{ij} = U(\psi)^{2/3}\delta_{ij}, \qquad \mathrm{d}V = U(\psi)\mathrm{d}^3x. \tag{6.1}$$

There is one physical held clock N. The variables ψ, ζ parameterize the clock and spatial response.

For material number per coordinate volume $\bar{n}$, the proper number density is $n = \bar{n}/U$. Let $\epsilon(n)$ be proper material energy density, with pressure $p = n\epsilon_n - \epsilon$. The static functional is

$$\boxed{W = \int \left[\frac{\zeta U}{F}\epsilon(\bar{n}/U) - \kappa \boldsymbol{\nabla}\psi \cdot \boldsymbol{\nabla}\zeta\right] \mathrm{d}^3x, \qquad \kappa = \frac{c^4}{2\pi G}.} \tag{6.2}$$

The fields approach $\psi = \zeta = 1$ at infinity. The weak source calibration is $F(1) = F'(1) = U(1) = 1$. Material support is included in ϵ and its physical work.

Equation (6.2) specifies a local static class: one isotropic spatial scale, quadratic first gradients, and affine dependence on the physical lapse and its gradient. It is broader than one guessed choice of F, U. Appendix A shows how a regular field redefinition brings a general member of that class into this form. A complementary lapse-gradient class is tested in Appendix B.

Variation with respect to ζ and ψ gives

$$\nabla^2\psi = -\frac{2\pi G}{c^4}\frac{U}{F}\epsilon, \tag{6.3}$$

$$\nabla^2\zeta = \frac{2\pi G}{c^4}\frac{\zeta U}{F}\left(\frac{F'}{F}\epsilon + \frac{U'}{U}p\right). \tag{6.4}$$

The pressure term appears because a change of ψ changes physical volume. Specifically, $\partial_\psi[U\epsilon(\bar{n}/U)] = -pU'$ at fixed $\bar{n}$.

At stationary fields, a small added local proper energy contributes $N_i \mathrm{d}E_i$; first-order field variations vanish. Eliminating the stationary fields thus leaves $H_{E_i} = N_i$ and reciprocal derivatives. Integrating the gradient term by parts with Eq. (6.3) also gives

$$H = W_{\text{stationary}} = -\kappa \lim_{\rho\to\infty} \oint \zeta\, \partial_\rho \psi \, \mathrm{d}A_0, \tag{6.5}$$

where $\mathrm{d}A_0$ is coordinate area. In spherical vacuum write

$$\psi = 1 + x, \quad \zeta = 1 - y, \quad x = \frac{GH}{2c^4\rho}, \quad y = \frac{GC}{2c^4\rho}. \tag{6.6}$$

The far clock is $N = 1 - G(H + C)/(2c^4\rho) + \cdots$, so its active charge is $(H + C)/2$. Complete-energy correspondence, Eq. (4.6), therefore requires $C = H$, or $y = x$. This has not specified the finite-radius burden profile or radial measure.

7 The source law selects the clock and radial response

7.1 Vary the source, its field and its physical support

For a supported shell, the interior fields are constant and the exterior solutions are Eq. (6.6). Define $\mathcal{L}(\psi) = U^{1/3}$, so the shell's physical radius is $R = \rho\mathcal{L}$. Evaluation of the exterior gradient integral gives the reduced functional

$$W = \frac{1 - y}{F} E + \frac{2\rho}{k} xy, \qquad k \equiv \frac{G}{c^4}. \tag{7.1}$$

Here $F, \mathcal{L}$ are evaluated at $\psi_s = 1 + x$. Three variations are needed. Varying y, x, and ρ, respectively, gives

$$H = \frac{2\rho x}{k} = \frac{E}{F}, \tag{7.2}$$

$$C = \frac{2\rho y}{k} = N\left[E\frac{F'}{F} + 2P_s\mathcal{A}_s\frac{\mathcal{L}'}{\mathcal{L}}\right], \tag{7.3}$$

$$2NP_s\mathcal{A}_s = Hy. \tag{7.4}$$

The material contributions in the last two equations follow from $\mathrm{d}E = -P_s\mathrm{d}\mathcal{A}_s$ as the physical radius changes. Omitting them would vary coordinate fields while holding the physical support work incorrectly fixed.

Substitute $C = H$, $y = x$, and Eq. (7.2). The remaining compatibility equation is

$$\boxed{(1 - x)\frac{\mathrm{d}\ln F}{\mathrm{d}x} + x\frac{\mathrm{d}\ln\mathcal{L}}{\mathrm{d}x} = 1.} \tag{7.5}$$

7.2 Apply the same source law across physical source surfaces

The actual source relation, Eq. (5.3), supplies

$$N^2 = 1 - \frac{4x}{\mathcal{L}}. \tag{7.6}$$

Use the same functions F, U across a continuous family of admitted spherical source surfaces. This universality supplies an interval of compactness values; a single boundary value would not determine a function. Set $a(x) = N_s$. Then

$$F = \frac{1 - x}{a}, \qquad \mathcal{L} = \frac{4x}{1 - a^2}. \tag{7.7}$$

Inserting these into Eq. (7.5) yields

$$\frac{\mathrm{d}a}{\mathrm{d}x} = \frac{a(1-a^2)}{(1+x)a^2-(1-x)}. \tag{7.8}$$

Inverting makes the equation linear in $x(a)$:

$$\frac{\mathrm{d}x}{\mathrm{d}a} = -\frac{1}{a} + x\frac{1+a^2}{a(1-a^2)}. \tag{7.9}$$

Its first integral is

$$\frac{x(1-a^2)}{a} - a - \frac{1}{a} = C_0. \tag{7.10}$$

The unloaded limit $x = 0, a = 1$ fixes $C_0 = -2$, giving $x(1-a^2) = (1-a)^2$. On the finite sourced branch,

$$\boxed{a = \frac{1-x}{1+x}, \qquad F = 1+x = \psi, \qquad U = (1+x)^6 = \psi^6.} \tag{7.11}$$

The other algebraic branch $a = 1$ at $x > 0$ requires infinite $\mathcal{L}$ in Eq. (7.6) and does not describe a finite sourced configuration.

The result selects the response functions on the interval sampled by the admitted source family. Extension beyond that interval uses an extended source family or an analytic continuation of the same branch; continuity alone would not select arbitrary unsampled strong-field behavior. The positive-lapse spherical exterior is the branch used here.

7.3 Convert to physical areal radius

With Eq. (7.11),

$$R = \rho(1+x)^2, \qquad \mathrm{d}\ell = (1+x)^2\mathrm{d}\rho, \qquad x = \frac{GH}{2c^4\rho}. \tag{7.12}$$

Because $\mathrm{d}R/\mathrm{d}\rho = 1 - x^2$,

$$\boxed{q_{RR} = \frac{(1+x)^2}{(1-x)^2} = \alpha^{-2}, \qquad \alpha^2 = 1 - \frac{2GH}{Rc^4} = 1 - \frac{2GM}{Rc^2}.} \tag{7.13}$$

The source law has selected the clock and radial responses together in the stated class. No orbital coefficient or Einstein field action was inserted into Eqs. (7.1)–(7.13). The resulting operational interval is the Schwarzschild exterior. That agreement is a result of this construction, rather than a claim to a new numerical spherical solution.

8 The work current is a consequence of the selected field

The held-clock relation already defines a direct differential object:

$$\boxed{\boldsymbol{\mathcal{J}} \equiv c^2\boldsymbol{D}\alpha = -\frac{\boldsymbol{D}(S^2)}{2\alpha}.} \tag{8.1}$$

From Eq. (3.14), $\boldsymbol{\mathcal{J}} = \alpha\boldsymbol{F}_{\mathrm{hold}}/m$. It is a clock-weighted work-current reading. The proper holding acceleration is $\boldsymbol{\mathcal{J}}/\alpha$. The field equations now determine the divergence of this same object.

For principal material stresses p_1, p_2, p_3, put $\operatorname{tr}\sigma = p_1 + p_2 + p_3$. In the selected conformal spatial variation, Eqs. (6.3)–(6.4) become

$$\nabla^2\psi = -\frac{2\pi G}{c^4}\psi^5\epsilon, \tag{8.2}$$

$$\nabla^2\zeta = \frac{2\pi G}{c^4}\zeta\psi^4(\epsilon + 2\operatorname{tr}\sigma). \tag{8.3}$$

For an isotropic fluid, $\operatorname{tr}\sigma = 3p$. Since $q_{ij} = \psi^4\delta_{ij}$ and $\alpha = \zeta/\psi$,

$$\boldsymbol{D}^2\alpha = \psi^{-6}\boldsymbol{\nabla}\cdot(\psi^2\boldsymbol{\nabla}\alpha) = \psi^{-6}(\psi\nabla^2\zeta - \zeta\nabla^2\psi). \tag{8.4}$$

It follows that

$$\boxed{\boldsymbol{D}\cdot\boldsymbol{\mathcal{J}} = \frac{4\pi G}{c^2}\alpha(\epsilon + \operatorname{tr}\sigma).} \tag{8.5}$$

At infinity $\alpha = 1 - GH/(c^4R) + \cdots$. Applying the divergence theorem to a complete isolated supported source gives

$$H = \int \alpha(\epsilon + \operatorname{tr}\sigma)\,\mathrm{d}V, \tag{8.6}$$

$$\boxed{\oint \boldsymbol{\mathcal{J}}\cdot\boldsymbol{n}\,\mathrm{d}A = \frac{4\pi GH}{c^2} = \frac{Q_{\mathrm{s}}^2}{6V_{\mathrm{ref}}}.} \tag{8.7}$$

Thus $\boldsymbol{D}\cdot\boldsymbol{\mathcal{J}} = 0$ in vacuum is an output of the field variation. Every complete vacuum reading surface encloses the same current charge. For the spherical exterior,

$$\mathcal{J}_R = c^2\alpha\frac{\mathrm{d}\alpha}{\mathrm{d}R} = \frac{GM}{R^2}. \tag{8.8}$$

The domain quantity Q_{s} and the work current express the same source charge in different units. The conserved combination in Eq. (8.7) is $Q_{\mathrm{s}}^2/V_{\mathrm{ref}}$. It does not imply that $4\pi R^2 S(R)$ remains the same number when the reference domain is changed with R.

8.1 Field energy in burden and current variables

Let $b = \ln\psi$. Rewriting the gradient part of Eq. (6.2) in physical volume gives

$$W_f = \int\left[\frac{|\boldsymbol{\mathcal{J}}|^2}{8\pi G\alpha} - \frac{c^4\alpha}{2\pi G}\left|\boldsymbol{D}b + \frac{\boldsymbol{\mathcal{J}}}{2c^2\alpha}\right|^2\right]\mathrm{d}V. \tag{8.9}$$

The spatial-response term is part of the same nonlinear energy. In the weak unloaded limit, writing $u_S = S^2/c^2$, the selected fields obey $\psi = 1 + u_S/4 + \cdots$, $\zeta = 1 - u_S/4 + \cdots$. Consequently

$$W_f = \frac{c^4}{32\pi G}\int|\boldsymbol{\nabla}u_S|^2\,\mathrm{d}^3x + \text{higher-order terms}. \tag{8.10}$$

The earlier weak gradient energy is retained inside the complete spatial construction.

On the selected spherical vacuum, $\psi = 2/(1+\alpha)$, and

$$\frac{\mathrm{d}W_f}{\mathrm{d}V} = \frac{|\boldsymbol{\mathcal{J}}|^2}{2\pi G(1+\alpha)^2}, \qquad W_{f,\mathrm{outside}} = H\frac{1-\alpha_s}{1+\alpha_s}. \tag{8.11}$$

For a thin shell, Eq. (5.5) gives $E = 2H/(1+\alpha_s)$, hence

$$W_{f,\mathrm{outside}} = E - H, \qquad H = \alpha_s E + W_{f,\mathrm{outside}}. \tag{8.12}$$

This identifies the energy contribution used in the worked example of Section 5. It is an on-solution decomposition, not an independently imposed field-energy density.

9 Perihelion from the selected energy and field

In the equatorial spherical exterior,

$$A = 1 - \frac{2GM}{Rc^2}, \qquad q_{RR} = A^{-1}, \quad q_{\varphi\varphi} = R^2. \tag{9.1}$$

The selected probe energy becomes

$$\mathcal{H} = \sqrt{A}\sqrt{m^2c^4 + c^2\left(Ap_R^2 + \frac{p_\varphi^2}{R^2}\right)}. \tag{9.2}$$

There is no explicit T or φ dependence. Hamilton's equations conserve $\mathcal{H}$ and p_φ. Define

$$\mathcal{E} = \frac{\mathcal{H}}{mc^2}, \qquad j = \frac{p_\varphi}{m}, \qquad \mu = GM. \tag{9.3}$$

Equation (3.8) gives $\beta = A/\mathcal{E}$, and the momentum relation gives

$$\frac{\mathrm{d}R}{\mathrm{d}\tau} = \frac{Ap_R}{m}, \qquad \frac{\mathrm{d}\varphi}{\mathrm{d}\tau} = \frac{j}{R^2}. \tag{9.4}$$

Squaring Eq. (9.2) and eliminating p_R yields

$$\left(\frac{\mathrm{d}R}{\mathrm{d}\tau}\right)^2 = c^2\left[\mathcal{E}^2 - A\left(1 + \frac{j^2}{c^2R^2}\right)\right]. \tag{9.5}$$

Set $w = 1/R$. Since $\mathrm{d}w/\mathrm{d}\varphi = -(1/j)\mathrm{d}R/\mathrm{d}\tau$,

$$\left(\frac{\mathrm{d}w}{\mathrm{d}\varphi}\right)^2 = \frac{c^2}{j^2}(\mathcal{E}^2 - 1) + \frac{2\mu w}{j^2} - w^2 + \frac{2\mu w^3}{c^2}. \tag{9.6}$$

Differentiate and extend by continuity through the turning points:

$$\boxed{\frac{\mathrm{d}^2w}{\mathrm{d}\varphi^2} + w = \frac{\mu}{j^2} + \frac{3\mu w^2}{c^2}.} \tag{9.7}$$

To first weak-field order, take the Kepler form $w_0 = (1 + e\cos\varphi)/p_\mathrm{o}$, with $p_\mathrm{o} = a_o(1 - e^2)$ and $j^2 = \mu p_\mathrm{o}$ at leading order. The resonant $\cos\varphi$ contribution of $3\mu w_0^2/c^2$ changes the radial oscillation to $\cos[(1 - \delta)\varphi]$, where $\delta = 3\mu/(p_\mathrm{o}c^2)$. Therefore

$$\boxed{\Delta\varphi = \frac{6\pi GM}{a_o(1 - e^2)c^2} = \frac{Q_\mathrm{s}^2}{4V_\mathrm{ref}a_o(1 - e^2)c^2} + \mathcal{O}\left(\frac{G^2M^2}{p_\mathrm{o}^2c^4}\right).} \tag{9.8}$$

The same source charge appears in the boundary law, work current and orbital advance. The $3\mu w^2/c^2$ term comes from the selected energy and radial response; it was not added as a target correction.

9.1 Why the orbit can precess while conserving energy

For fixed E_0, Eq. (3.6) gives along the stationary free orbit

$$\Delta(p_iv^i) = -E_0\Delta\beta. \tag{9.9}$$

Internal progression and translation are coupled through one conserved generator. Near perihelion the clock-weighted term decreases while the momentum–velocity term increases. The ideal clock's

local internal energy can remain constant throughout. At low speed $p_i v^i = 2K + \cdots$, so this term is not an additional kinetic-energy reservoir.

After a radial cycle, the body returns to its perihelion radius with the same energy, speed and clock rate. Its direction has advanced because radial recurrence and angular motion accumulate at different rates. No secular increase in orbit size or kinetic energy is required.

This connection can be written as an exact orbital identity. Define the radial action $I_r = (2\pi)^{-1} \oint p_R \mathrm{d}R$. Implicit differentiation of the energy surface gives

$$\frac{\partial I_r}{\partial \mathcal{H}} = \frac{T_r}{2\pi}, \qquad \frac{\partial I_r}{\partial p_\varphi} = -\frac{\Phi_r}{2\pi}, \qquad \frac{\partial I_r}{\partial E_0} = -\frac{\tau_r}{2\pi}. \tag{9.10}$$

Consequently,

$$\frac{\partial \Phi_r}{\partial E_0} = \frac{\partial \tau_r}{\partial p_\varphi}. \tag{9.11}$$

The angle and accumulated clock progression are derivatives of the same orbital action. Related redshift–energy first laws have an established relativistic formulation [4].

For the selected eccentric weak-field branch, averaging $\beta = 1 - \mu/(Rc^2) - v^2/(2c^2) + \cdots$ over a leading Kepler cycle uses $\langle 1/R \rangle = 1/a_o$ and $\langle v^2 \rangle = \mu/a_o$. Thus

$$1 - \frac{\tau_r}{T_r} = \frac{3\mu}{2a_o c^2} + \mathcal{O}(c^{-4}), \qquad \Delta\varphi = \frac{4\pi}{1-e^2}\left(1 - \frac{\tau_r}{T_r}\right) + \mathcal{O}(c^{-4}). \tag{9.12}$$

This is a consequence of the solved branch, rather than a second independent selection of the field.

9.2 A Mercury-scale calculation

Use the solar gravitational parameter and approximate J2000 orbital elements published by JPL [1, 2]:

$$\mu_\odot = 1.32712440041279419 \times 10^{20}\ \mathrm{m^3\,s^{-2}}, \tag{9.13}$$

$$a_o = 0.38709927\ mau, \qquad e = 0.20563593, \tag{9.14}$$

$$1\ mau = 149597870700\ mm, \qquad c = 299792458\ \mathrm{m\,s^{-1}}. \tag{9.15}$$

For exact quadrature use a radial phase χ, distinct from azimuth:

$$R = \frac{p_\mathrm{o}}{1 + e\cos\chi}, \qquad \eta_o = \frac{\mu}{p_\mathrm{o} c^2}. \tag{9.16}$$

Solving the two turning-point equations gives

$$j^2 = \frac{\mu p_\mathrm{o}}{1 - \eta_o(3 + e^2)}, \tag{9.17}$$

$$\mathcal{E}^2 = \frac{(1-2\eta_o)^2 - 4e^2\eta_o^2}{1 - \eta_o(3+e^2)}, \tag{9.18}$$

$$\frac{\mathrm{d}\varphi}{\mathrm{d}\chi} = \left[1 - \eta_o(6 + 2e\cos\chi)\right]^{-1/2}, \tag{9.19}$$

$$\frac{\mathrm{d}\tau}{\mathrm{d}\chi} = \frac{R^2}{j}\frac{\mathrm{d}\varphi}{\mathrm{d}\chi}, \qquad \frac{\mathrm{d}T}{\mathrm{d}\chi} = \frac{\mathcal{E}}{A}\frac{\mathrm{d}\tau}{\mathrm{d}\chi}. \tag{9.20}$$

Integration over $0 \le \chi \le 2\pi$ gives the results in Table 1. The classical comparison has the same source and areal turning radii. These are isolated-source calculations using measured source inputs and fitted approximate orbital elements, not a new fit to Mercury ranging data. Other planetary perturbations and solar multipoles are outside this illustration.

Table 1: Mercury-scale orbital and clock calculations. The angle is converted to arcseconds per Julian century using the corresponding radial period. Rounded values retain the physical scale without implying an observational uncertainty at the last digit.

Quantity	Calculated value
Leading spherical perihelion advance	42.98 arcsec/century
Exact spherical quadrature	42.98 arcsec/century
Clock deficit $T_r - \tau_r$	0.290710 s/cycle
Mean clock deficit $1 - \tau_r/T_r$	3.82484×10^{-8}
Reference radial-period excess over the comparison	0.581419 s/cycle

9.3 The same response as a propagation delay

For the radial null capacity branch $\beta = 0$, the selected $q_{RR} = \alpha^{-2}$ gives

$$\left|\frac{\mathrm{d}R}{\mathrm{d}T}\right| = c\alpha^2, \qquad n_R = \alpha^{-2}. \tag{9.21}$$

Since $\mathrm{d}\ell = |\mathrm{d}R|/\alpha$ and $\mathrm{d}t_h = \alpha \mathrm{d}T$, a held local observer measures $\mathrm{d}\ell/\mathrm{d}t_h = c$. The radial delay in reference time and the spatial-measure representation encode the same operational result. This concerns light propagation in the selected stationary field.

10 One energy for moving sources and leading rotation

The static selection can be extended to conservative source motion through the first relativistic correction. The extra physical requirement here is explicit: the complete interacting system must admit the same Lorentz observer-change relations as an unloaded capacity-governed system. Free-particle dispersion alone would not impose that requirement on interactions.

10.1 Leave the mixed-momentum coefficients unknown

For two source cells with absorbed self energies m_1c^2, m_2c^2, put $r = |\boldsymbol{x}_1 - \boldsymbol{x}_2|$ and $\boldsymbol{n} = (\boldsymbol{x}_1 - \boldsymbol{x}_2)/r$. Expand

$$H = (m_1 + m_2)c^2 + H_N + \frac{H_{1\mathrm{PN}}}{c^2} + \mathcal{O}(c^{-4}), \tag{10.1}$$

where

$$H_N = \frac{p_1^2}{2m_1} + \frac{p_2^2}{2m_2} - \frac{Gm_1m_2}{r}. \tag{10.2}$$

Expansion of the selected static common energy with local matter momentum gives

$$H_{1,s} = -\frac{p_1^4}{8m_1^3} - \frac{p_2^4}{8m_2^3} - \frac{3G}{2r}\left(\frac{m_2p_1^2}{m_1} + \frac{m_1p_2^2}{m_2}\right) + \frac{G^2m_1m_2(m_1+m_2)}{2r^2}. \tag{10.3}$$

The algebra producing these coefficients is displayed in Appendix D. In the conservative parity-even pair class, retain the two leading mixed scalars with unknown coefficients:

$$H_{1\mathrm{PN}} = H_{1,s} + \frac{G}{2r}\left[A_m \boldsymbol{p}_1 \cdot \boldsymbol{p}_2 + B_m(\boldsymbol{n}\cdot\boldsymbol{p}_1)(\boldsymbol{n}\cdot\boldsymbol{p}_2)\right]. \tag{10.4}$$

The first spatial moment of the weak common energy gives the boost centroid $\boldsymbol{\mathcal{G}} = \boldsymbol{\mathcal{G}}_0 + c^{-2}\boldsymbol{\mathcal{G}}_1 + \cdots$, with

$$\boldsymbol{\mathcal{G}}_0 = m_1\boldsymbol{x}_1 + m_2\boldsymbol{x}_2, \tag{10.5}$$

$$\boldsymbol{\mathcal{G}}_1 = \sum_{i=1}^{2} \boldsymbol{x}_i \frac{p_i^2}{2m_i} - \frac{Gm_1m_2}{2r}(\boldsymbol{x}_1 + \boldsymbol{x}_2). \tag{10.6}$$

The clock-weighted source terms contribute $-Gm_1m_2(\boldsymbol{x}_1+\boldsymbol{x}_2)/r$. The cross-gradient field energy is positive, centered at the midpoint, and contributes half that magnitude back. This fixes the displayed centroid convention.

The first correction to the boost–energy relation is

$$\{\boldsymbol{\mathcal{G}}_0, H_{1\mathrm{PN}}\} + \{\boldsymbol{\mathcal{G}}_1, H_N\} = 0. \tag{10.7}$$

Evaluate the three-dimensional brackets before aligning the separation with an axis. The residual components then have the form

$$\mathcal{B}_\perp = \frac{G}{2r}(m_2 p_{1\perp} + m_1 p_{2\perp})(A_m - 7), \tag{10.8}$$

$$\mathcal{B}_\parallel = \frac{G}{2r}(m_2 p_{1\parallel} + m_1 p_{2\parallel})(A_m + B_m - 8). \tag{10.9}$$

They vanish for arbitrary momenta only if

$$\boxed{A_m = 7, \quad B_m = 1, \quad \Delta H = \frac{G}{2rc^2}[7\boldsymbol{p}_1\cdot\boldsymbol{p}_2 + (\boldsymbol{n}\cdot\boldsymbol{p}_1)(\boldsymbol{n}\cdot\boldsymbol{p}_2)].} \tag{10.10}$$

This is the established first-post-Newtonian two-body structure in this canonical convention [3]. The coefficients are selected here by bulk consistency, without using a perihelion target.

For a circular orbit, with $M_b = m_1 + m_2$, $m_r = m_1m_2/M_b$, $\nu = m_r/M_b$, and $x_\Omega = (GM_b\Omega/c^3)^{2/3}$, the energy gives

$$\frac{E_{\mathrm{bind}}}{m_r c^2} = -\frac{x_\Omega}{2} + \left(\frac{3}{8} + \frac{\nu}{24}\right) x_\Omega^2 + \mathcal{O}(x_\Omega^3). \tag{10.11}$$

Keeping only the instantaneous static extension, $A_m = B_m = 0$, instead gives $3/8 - 9\nu/8$ for the quadratic coefficient. Both share the test-mass limit; finite mass ratio distinguishes them. Clock derivatives alone do not detect the mixed term because it has no explicit bare-mass dependence at fixed canonical momenta. Complete-state consistency must address momentum as well as internal clocks.

10.2 The selected kernel has a rotational consequence

Let $\boldsymbol{j}_p$ be canonical source momentum per coordinate volume and $\boldsymbol{w}$ its transport response. The Green tensor of $-\nabla^2 - \lambda\boldsymbol{\nabla}\boldsymbol{\nabla}\cdot$ is

$$G_{ij}(\boldsymbol{r}) = \frac{(\lambda+2)\delta_{ij} + \lambda n_i n_j}{8\pi(1+\lambda)r}. \tag{10.12}$$

The selected tensor ratio in Eq. (10.10) is $7:1$, hence $\lambda/(\lambda+2) = 1/7$ and $\lambda = 1/3$. Its normalized local representation is

$$-\nabla^2\boldsymbol{w} - \frac{1}{3}\boldsymbol{\nabla}(\boldsymbol{\nabla}\cdot\boldsymbol{w}) = \frac{16\pi G}{c^2}\boldsymbol{j}_p. \tag{10.13}$$

The shared constrained energy is

$$W_w = \int \left[\boldsymbol{j}_p\cdot\boldsymbol{w} - \frac{c^2}{32\pi G}\left(|\boldsymbol{\nabla}\boldsymbol{w}|^2 + \frac{1}{3}(\boldsymbol{\nabla}\cdot\boldsymbol{w})^2\right)\right] \mathrm{d}^3x. \tag{10.14}$$

At its stationary solution, $H_w = \frac{1}{2}\int \boldsymbol{j}_p \cdot \boldsymbol{w}\mathrm{d}^3x$, which counts each pair interaction once.

For a localized steady current in its rest frame, $\boldsymbol{\nabla}\cdot\boldsymbol{j}_p = 0$ and its net momentum vanishes. Then

$$\boldsymbol{w}(\boldsymbol{x}) = \frac{4G}{c^2}\int \frac{\boldsymbol{j}_p(\boldsymbol{x}')}{|\boldsymbol{x}-\boldsymbol{x}'|}\mathrm{d}^3x'. \tag{10.15}$$

Stationarity makes the first current moment antisymmetric. With $\boldsymbol{J}_{\mathrm{ang}} = \int \boldsymbol{x}' \times \boldsymbol{j}_p \mathrm{d}^3x'$, the leading rotational term is

$$\boxed{\boldsymbol{w} = \frac{2G}{c^2 r^3}\boldsymbol{J}_{\mathrm{ang}} \times \boldsymbol{x}, \qquad \boldsymbol{\Omega}_{\mathrm{tr}} = \frac{2G\boldsymbol{J}_{\mathrm{ang}}}{c^2 r^3}.} \tag{10.16}$$

Thus the leading angular transport rate falls as r^{-3}, with its coefficient inherited from the selected interaction. This is a weak-field conservative rotational response. Equation (10.13) is an elliptic representation of that kernel, not a field radiation equation or a complete rotating-black-hole solution.

11 Checks of the construction

The checks examine more than reproduction of one orbit. Alternative responses were retained through their own source and orbital calculations. The source requirement can therefore reject a response that passes an energy test, and complete-energy accounting can reject a deformation that passes an exterior test.

Table 2: Representative checks retained from the research. Numerical residuals are implementation checks in dimensionless test units, not measurement uncertainties.

Check	Requirement examined	Result
Nested supported shells	One energy, clock derivatives, exterior charge	600 generated systems; charge/energy residual about 2×10^{-16}.
Continuous supported sources	Local energy additions read the local clock	Twelve window variations; maximum clock-derivative difference 1.3×10^{-10}.
Integrated source current	Complete energy includes stress	Agreement about 6×10^{-13} in the checked stellar family.
Alternative static responses	Source boundary and physical orbital angle	Energy-consistent alternatives give different source residuals and different advances; Appendix C.
Mixed source momenta	Transverse and longitudinal boost consistency	Exact symbolic selection $A_m = 7, B_m = 1$.
Steady rotating ring	Selected kernel and angular far field	Direct 32,768-element integration approaches the dipole response.

The continuum spherical checks use

$$B = 1 - \frac{2Gh(R)}{Rc^4}, \qquad h' = 4\pi R^2\epsilon, \tag{11.1}$$

$$\frac{N'}{N} = \frac{G[h + 4\pi R^3 p]}{R^2 c^4 B}, \qquad p' = -(\epsilon + p)\frac{N'}{N}. \tag{11.2}$$

The proper volume is $4\pi R^2 \mathrm{d}R/\sqrt{B}$. These are the familiar spherical equilibrium relations [5]; independent conformal-field integrations were compared with them. Inside matter, N need not equal $\sqrt{B}$. Exterior source relations must not be applied as local interior identities.

One synthetic source with $G = c = 1$ and equation of state $\epsilon = n + n^2, p = n^2$, at central $n = 0.3$, has $H = 0.163627672143$, proper material energy 0.203251778534, and surface clock 0.760621015079. Its spherical binding contribution is -0.039624106392. The two decompositions

$$H = E_{\text{proper}} + W_{\text{binding}} = E_{\text{redshifted}} + W_f \tag{11.3}$$

agree: the sign of a named field contribution depends on which material energy it complements. This is not a claim of a unique invariant local gravitational energy density.

These synthetic checks verify equations and their mutual compatibility. The Mercury calculation uses published source inputs but is not an independent observational discrimination between SET and GR. The paper's result is the connected selection and its consequences.

12 Field memory and the complete state

The common-energy requirement allows memory. If two histories leave the same present material sources but different field configurations, they have left different complete states. They can have different energies or the same total energy distributed differently. Neither possibility violates Eq. (4.2).

A separate causal scalar control in the research demonstrated conserved total energy with history-dependent field states. A longer run transported the local history difference out of the source neighborhood without adding a decay term. This establishes an accounting distinction: conservation does not require a permanent local burden pattern at an old source position. The control has its own supplied kinetic term and is not the selected static field theory of this paper. Its role is to clarify the meaning of a complete state for a future causal extension.

The natural continuation is one positive causal field evolution whose stationary limit is Eq. (6.2) with the selected response, and whose conservative momentum limit is Eq. (10.10). The source, clock and energy requirements developed here provide concrete checks for that continuation.

13 What the construction establishes

SET's capacity interpretation becomes dynamically restrictive when the same energy must generate material motion, internal clock progression and the response of the complete source. Within the local classes examined here, these requirements and the SET source law select a consistent spherical field, its conserved work current and its perihelion dynamics.

The selections are connected. Actual clock progression fixes the probe coupling. Complete clock universality links the source charge to total energy. Physical support work and the source law constrain the spatial response. The resulting field equations supply current conservation. Applying complete-system consistency to bulk motion then selects the leading mixed-momentum interaction. Several equations arise from a small set of physical requirements, rather than from a separate orbital fit at each step.

The field content and derivative classes remain explicit parts of the construction. The known spherical and leading two-body formulas are the checks recovered by this route. The result is a concrete way for SET's capacity and source principles to participate in selecting those formulas, with each mathematical connection available for inspection.

A Normal form of the static energy class

Let q be a scalar spatial-response variable. A regular local isotropic quadratic first-gradient energy affine in $N, \boldsymbol{\nabla} N$ has the form

$$W_f = \int [Na(q)|\boldsymbol{\nabla} q|^2 + b(q)\boldsymbol{\nabla} N \cdot \boldsymbol{\nabla} q] \mathrm{d}^3 x. \tag{A.1}$$

Set $\zeta = NF(\psi)$ and choose a monotone $\psi(q)$. Matching this expression to $-\kappa \boldsymbol{\nabla}\psi \cdot \boldsymbol{\nabla}\zeta$ requires

$$F = -\frac{b}{\kappa \psi_q}, \qquad \frac{\psi_{qq}}{\psi_q} = \frac{b_q - a}{b}. \tag{A.2}$$

These equations have a local regular solution where $b \neq 0$ and the field redefinition is nonsingular. The remaining freedom is carried by the material functions F, U, which are selected in Section 7. The normal-form statement does not include arbitrary lapse-gradient squares or higher derivatives; the next appendix tests a complementary class.

B A complementary lapse-gradient test

Fix the conformal volume $\mathrm{d}V = \psi^6 \mathrm{d}^3 x$ and consider

$$W_f = \kappa \int \left[a_K N |\boldsymbol{\nabla}\psi|^2 + b_K \psi \boldsymbol{\nabla} N \cdot \boldsymbol{\nabla}\psi + d_K \frac{\psi^2}{N} |\boldsymbol{\nabla} N|^2 \right] \mathrm{d}^3 x. \tag{B.1}$$

This homogeneous class can also be motivated by requiring the same local energy functional under a constant physical clock and ruler calibration, with no separate preferred ruler field. For general coefficient functions $a(\psi), b(\psi), d(\psi)$, that requirement gives

$$a' = 0, \qquad \psi b' = b, \qquad \psi d' = 2d, \tag{B.2}$$

which produces the constants and powers in Eq. (B.1). This is a specified operational covariance requirement in addition to the choice of local first-gradient terms.

For asymptotic energy charges P, E_a, write $\psi = 1 + GP/(2c^4\rho) + \cdots$ and $N = 1 - GE_a/(c^4\rho) + \cdots$. The field and support identities give

$$H = -b_K P + 4 d_K E_a, \qquad a_K P = b_K E_a. \tag{B.3}$$

Complete energy and the common weak source calibration require $-b_K^2/a_K + 4d_K = 1$. Expanding the actual source-boundary relation then gives at second order $a_K = b_K = 4d_K - 1$. At cubic order its residual is

$$-\frac{4d_K}{3(4d_K - 1)} \left(\frac{GE_a}{\rho c^4} \right)^3. \tag{B.4}$$

Consequently $d_K = 0$, $a_K = b_K = -1$, reproducing the selected cross-gradient energy. The degenerate case $d_K = 1/4$, $a_K = b_K = 0$, has no nonzero weak spherical vacuum gradient. This test uses more information than leading Mercury precession: the remaining nonzero d_K cases share the leading weak orbital coefficient but fail the exact source relation. The two classes tested here overlap; neither is a classification of every possible field theory.

C Controls and the role of each requirement

The scalar capacity equality alone permits any admissible path if β is assigned from that path. Requiring the actual internal Hamiltonian to generate β adds dynamical content. Test-body

additivity supplies the Euler identity used to select the complete probe coupling. Removing that condition admits additional clock-consistent energy functions. The source response is therefore tested separately, rather than folded into the probe proof.

A useful family of static controls is

$$F = [1 - \lambda(\psi - 1)]^{-1/\lambda}, \qquad U = F^{3(1-\lambda)}, \tag{C.1}$$

with the exponential limit at $\lambda = 0$. These satisfy the complete-energy/support compatibility equation over their regular domains. The physical source-boundary condition selects $\lambda = -1$, which gives $F = \psi, U = \psi^6$. The leading perihelion advances of the other members, relative to the selected result, are

$$\frac{\Delta\varphi_\lambda}{\Delta\varphi_{-1}} = \frac{17 - 7\lambda}{24} + \text{higher-order terms.} \tag{C.2}$$

Their differing angles are computed using their own physical radial responses. They are not just coordinate substitutions of the same orbit.

Complete interior accounting adds another test. A deformation can be constructed to vanish on the selected spherical vacuum while contributing inside extended material. The examined deformation fails the equality between complete energy and exterior charge in supported stars. Exterior agreement alone would miss it. These controls identify the independent work done by the clock, complete-energy and source requirements.

D Expansion supplying the static two-source coefficients

Use $\varepsilon_c = c^{-2}$ as a formal expansion parameter and write the cell fields as

$$\psi_i = 1 + \varepsilon_c a_i + \varepsilon_c^2 b_i + \cdots . \tag{D.1}$$

For two cells with self energies absorbed into m_i, the selected source equation and local matter energy give

$$a_1 = \frac{Gm_2}{2r}, \qquad a_2 = \frac{Gm_1}{2r}, \tag{D.2}$$

$$b_1 = \frac{G}{2r}\left(\frac{p_2^2}{2m_2} - m_2 a_2\right), \qquad b_2 = \frac{G}{2r}\left(\frac{p_1^2}{2m_1} - m_1 a_1\right). \tag{D.3}$$

The complete mass-equivalent energy in the static cell extension is

$$\frac{H}{c^2} = \sum_i \frac{1}{\psi_i}\sqrt{m_i^2 + \varepsilon_c \frac{p_i^2}{\psi_i^4}}. \tag{D.4}$$

Expand each term:

$$\frac{1}{\psi_i}\sqrt{m_i^2 + \varepsilon_c p_i^2/\psi_i^4} = m_i + \varepsilon_c\left(\frac{p_i^2}{2m_i} - m_i a_i\right) \tag{D.5}$$

$$+ \varepsilon_c^2\left[m_i(a_i^2 - b_i) - \frac{5a_i p_i^2}{2m_i} - \frac{p_i^4}{8m_i^3}\right] + \cdots . \tag{D.6}$$

Summing and substituting a_i, b_i gives H_N and Eq. (10.3). This calculation supplies the instantaneous static contribution. Section 10 then determines the mixed-momentum contribution by complete-system bulk consistency.

E Quantum clocks and phase progression

The selected probe energy has a direct quantum-clock connection when standard quantum evolution is supplied. Add a small internal Hamiltonian h to mc^2. At fixed canonical momentum,

$$\mathcal{H}(mc^2 + h) = \mathcal{H}(mc^2) + \beta h + \mathcal{O}(h^2). \tag{E.1}$$

In a weak field, with $\Phi = -S^2/2$,

$$H_{\text{clock,ref}} = h\left(1 + \frac{\Phi}{c^2} - \frac{P^2}{2m^2c^2}\right) + \cdots . \tag{E.2}$$

This agrees with the familiar relativistic coupling of internal clocks and motion [6]. For a prescribed classical path and a fixed local h,

$$i\hbar\frac{\mathrm{d}|\chi_c\rangle}{\mathrm{d}T} = \beta(T)h|\chi_c\rangle, \qquad |\chi_c(T)\rangle = e^{-ih\tau(T)/\hbar}|\chi_c(0)\rangle. \tag{E.3}$$

Internal phase rates change while the local energy distribution remains constant. With the semiclassical identifications $\mathcal{H} = \hbar\omega$, $p_i = \hbar k_i$, Eq. (3.6) becomes

$$\omega - k_i v^i = \frac{E_0}{\hbar}\beta. \tag{E.4}$$

For an observable internal clock one uses relative phases between levels; a single overall rest-energy phase is not itself a clock. These relations give a quantum-language version of the same clock–motion balance. They use quantum mechanics as an additional established framework, not as an output of the scalar capacity relation.

F Reproducibility and derivation boundaries

The accompanying research records retain symbolic checks, numerical integrations and alternative-response controls. The paper's derivation is self-contained; those files provide executable verification. The principal records are the capacity-evolution, clock–inertia, motion-redistribution, interacting-source, and common-energy investigations of 17–18 October 2026. The final common-energy archive contains the following focused checks:

Script	What it checks
`check_general_spatial_selection.py`	Shell variations, first integral, selected F, U and radial response.
`check_native_current_energy.py`	Current divergence and burden/current energy identities.
`check_selected_orbit.py`	Exact orbit equation after selection.
`check_interacting_momentum_selection.py`	Static expansion, boost brackets, coefficients and circular energy.
`check_selected_transport_field.py`	Vector kernel, normalization and angular far field.

The spherical continuum and moving-shell calculations verify a selected branch; they are not additional independent proofs of the static selection. The separate causal memory control is not merged into that branch. The present isotropic calculation concerns spherical sources and the stated leading momentum extension; SET's collective galaxy-area and cosmological applications are separate calculations.

References

[1] NASA/JPL Solar System Dynamics, *Astrodynamic Parameters*. https://ssd.jpl.nasa.gov/astro_par.html. Accessed 18 October 2026.

[2] NASA/JPL Solar System Dynamics, *Approximate Positions of the Planets*, Table 1: J2000 elements fitted for 1800–2050. https://ssd.jpl.nasa.gov/planets/approx_pos.html. Accessed 18 October 2026.

[3] T. Damour, P. Jaranowski and G. Schäfer, "Poincaré invariance in the ADM Hamiltonian approach to the general relativistic two-body problem," *Physical Review D* **62**, 021501 (2000); erratum **63**, 029903 (2001). https://arxiv.org/abs/gr-qc/0003051.

[4] A. Le Tiec, L. Blanchet and B. F. Whiting, "The First Law of Binary Black Hole Mechanics in General Relativity and Post-Newtonian Theory," *Physical Review D* **85**, 064039 (2012). https://arxiv.org/abs/1111.5378.

[5] S. Gao, "A general maximum entropy principle for self-gravitating perfect fluid," *Physical Review D* **84**, 104023 (2011). https://arxiv.org/abs/1109.2804.

[6] A. R. H. Smith and M. Ahmadi, "Quantum clocks observe classical and quantum time dilation," *Nature Communications* **11**, 5360 (2020). https://doi.org/10.1038/s41467-020-18264-4.